



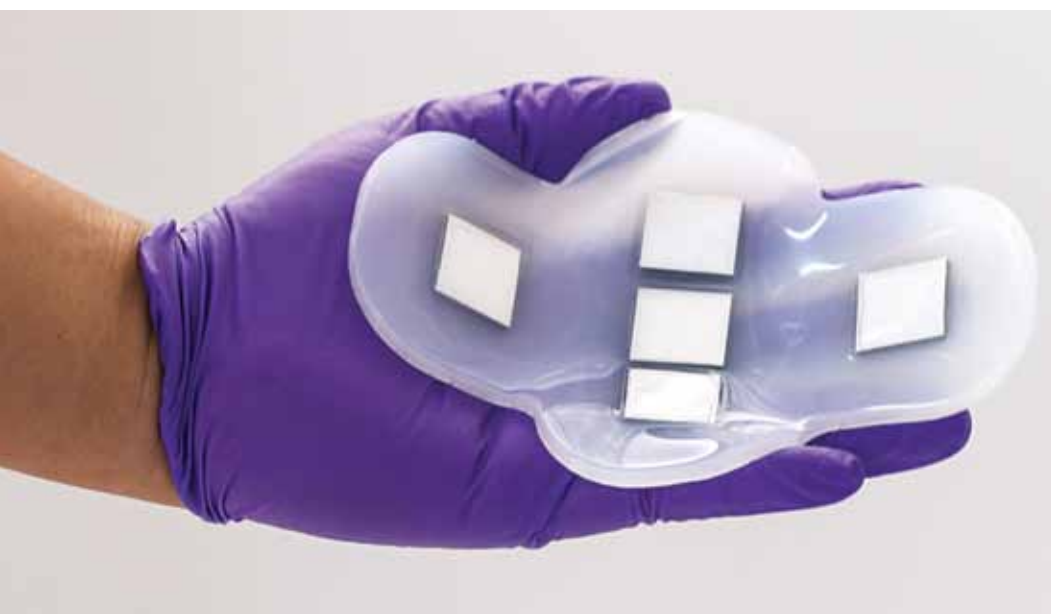
RESEARCH & INNOVATION UPDATES

Wearable ultrasound patch for advanced organ imaging

MIT researchers have unveiled a wearable ultrasound patch capable of non-invasively imaging internal organs, explicitly targeting the bladder, without the need for an operator or ultrasound gel. This technology offers precise

visualisation of bladder fullness, presenting advantages for individuals grappling with bladder or kidney afflictions. Unlike cumbersome traditional ultrasound probes, this patch adheres comfortably to the skin, forged from pliant

silicone rubber. It boasts five ultrasound arrays of piezoelectric substance, forming a cross-shaped configuration. With dimensions of approximately 12 by 8 centimetres, it captures the entire bladder area and sticks securely beneath clothing due to its innate adhesive properties. The team is developing a smart-phone-sized portable device for on-the-go image viewing to adapt the technology for imaging other organs like the pancreas, liver, or ovaries, which will involve adjusting ultrasound frequencies and creating new piezoelectric materials.



Wearable ultrasound patch (Credit: Photo courtesy of the researchers)

Fast-charging lithium-ion batteries with graphite anodes

Researchers at Huazhong University of Technology in China have developed a novel approach for creating fast-charging lithium-ion batteries (LiBs) using a graphite-based material. The team conducted experiments to understand the influence of solid electrolyte interface (SEI) components on the Li^+ solvation structure, a key factor in reducing battery charge time. They identified a material combination that improves the Li^+ desolvation process, enabling quicker Li^+ ion movement across the SEI. Building on this, they turned their attention to

a promising anode material called P-S graphite, composed of an ultra-thin phosphorus layer atop a graphite base. By manufacturing anodes from P-S graphite and integrating them into lithium-ion battery cells, they could assess the material's practical performance. The methodology they developed and their promising experimental results have the potential to significantly advance the development of faster-charging and longer-lasting LiBs, addressing the escalating demands of the electronics industry.